



**UNIVERSITI
MALAYA**

WID3002 NATURAL LANGUAGE PROCESSING

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**GROUP PROJECT:
TOXIC COMMENT DETECTOR**

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GROUP 8

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1.0 Introduction

With the rapid growth of social media and online discussion platforms, user-generated content (UGC) has become a necessity for digital communication, with billions of posts, tweets, and comments shared daily across major platforms such as Facebook, X and Reddit. However, the global advocacy for free speech allows individuals to express their thoughts uninhibitedly across these spaces. Consequently, even with rigorous traffic filtering, online platforms are not immune to massive volumes of toxic comments, including hate speech, threats, and offensive language. Such harmful content negatively affects users, reduces the quality of online interaction, and creates unsafe digital environments that promote a rude community. For instance, research shows that approximately 13% of 2 billion analyzed comments on Reddit were toxic, with toxic behavior dynamically adapting across different digital communities.

To mitigate this, this project presents the development of a Toxic Comment Detector, an intelligent text classification system that automatically identifies whether a comment is toxic or non-toxic. By utilizing a supervised machine learning approach, the system deploys a pretrained multilingual transformer model, XLM-RoBERTa, to analyze tweets and comments with high accuracy. This architecture enables the system to capture contextual nuance, slang, and cultural references across texts in English, Malay or a mix of both. The completed system features a modular backend built on Python, PyTorch, and HuggingFace Transformers, integrated with a clean, web-based Gradio frontend interface that provides users with instant predictions alongside confidence scores.

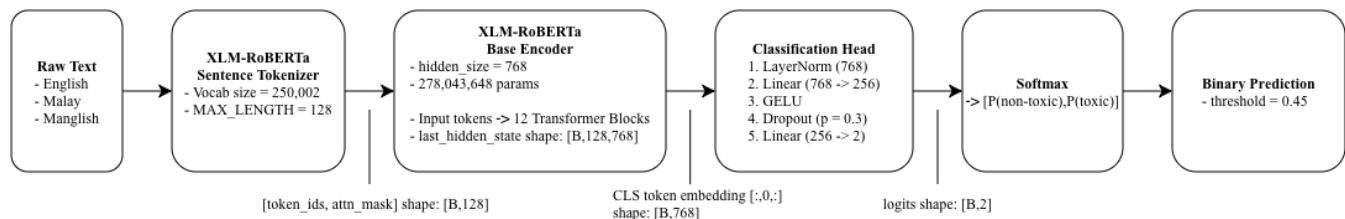
1.1 Problem Statement

Online platforms like Reddit, Facebook, and X receive massive amounts of user-generated data every single day. Due to widespread smartphone usage and the sheer velocity of incoming data, relying on manual content moderation has become very impractical and unscalable. In 2023, the Malaysian Communications and Multimedia Commission (MCMC) reported a massive increase in harmful content on social media platforms like TikTok, reaching over 25 thousand cases, which includes scams, fake news, and hate speech (BERNAMA, 2023). When left unchecked, these toxic contents might persist on social media channels, causing emotional harm to users and damaging the overall integrity of online platforms.

Developing a Natural Language Processing (NLP) solution for this problem poses some technical challenges. Social media commentary in multicultural regions like Malaysia is highly informal and bilingual, primarily consisting of English and Malay text. Traditional text classification approaches, such as Bag-of-Words (BOW) and Word2Vec, have always been unable to analyze the semantic meaning and context of social media language, which commonly includes local Malay slang, English abbreviations, misspellings, informal sentence structures, and context-dependent insults. Furthermore, standard monolingual transformer models lack the ability to switch between English and Malay text without losing contextual accuracy. Therefore, there is a critical need for the development of an intelligent multilingual language detector that can contextually analyze and filter out toxic language from social media streams.

2.0 Solution Architecture

Our solution is a transfer-learning architecture based on XLM-RoBERTa (xlm-roberta-base), a multilingual transformer pre-trained on 2.5TB of CommonCrawl text across 100 languages. Its shared subword representations across English and Malay allow generalisation to code-mixed Manglish without needing a separate model per language. The end-to-end pipeline spans five stages: dataset preparation, tokenisation and encoding, training and classifier engineering, evaluation and Gradio-based deployment. The architecture is as shown below:



Multilingual text is tokenised via SentencePiece into subword IDs, padded to 128 tokens, encoded through a 12-layer transformer of hidden size 768 and the [CLS] token embedding is passed through a custom two-layer classification head to produce a binary toxic or non-toxic prediction.

2.1 Solution Design

2.1.1 Tokenisation and Encoding

The XLMRobertaTokenizer uses SentencePiece byte-pair encoding over a 250,002 token vocabulary. An empirical token length analysis confirmed 95th percentile at 61 subword tokens and 100 as maximum, thus MAX_LENGTH=128 is enough to cover all training samples without truncation and buffer for longer Manglish inputs. A custom ToxicComment dataset handles adaptive tokenisation across 6 DataLoaders, including train, validation, combined-test and per-language test sets. The training loader is configured with shuffle=True, num_worker=2 and pin_memory=True for an accelerated GPU data transfer.

Next, 3 pooling strategies were benchmarked, which are CLS, mean and max. CLS pooling is finally selected because pre-training of XLM-RoBERTa explicitly optimises the <s> token as a whole-sequence summary, which is the natural input for a downstream classification head.

2.1.2 Training and Classification

The ClassificationHead design mirrors RoBERTa internal conventions, which are normalisation convention (LayerNorm) and activation function (GELU) with Xavier uniform initialisation to prevent unstable gradients at the beginning of fine tuning. LayerNorm is applied first to stabilise the distribution of the CLS embeddings, which can change in scale across the different languages and sample comments. The intermediate Linear layer compresses the 768-dimensional embedding to 256 dimensions with Dropout (p=0.3) applied before the final projection to reduce overfitting. The ToxicClassifier combines the XLM-RoBERTa encoder and the classification head into a single end-to-end trainable module, with the CLS token embedding as the interface between the two components and outputs raw logits over two classes without Softmax as CrossEntropyLoss will automatically apply it internally during training.

2.2 Dataset Overview

This project primarily uses the Bilingual Malay-English Social Media Dataset for Binary Hate Speech Detection published by Jun Chen Tan & Lee-Yeng Ong on Mendeley Data. This dataset contains 26,985 bilingual English and Malay social media posts collected from five publicly available sources, HateXplain, HateM, Toxicity-Small, Snapshot-Twitter-2022 and Supervised-Twitter specifically for binary hate speech detection tasks. As the dataset is fairly balanced across classes with 14,642 non-hate and 12,343 hate-labeled entries, and across language with 13,609 English and 13,376 Malay texts, it gives the model equal exposure to both languages and to toxic and non-toxic content during training, making it fair and suitable for this project.

The project uses a manually curated set of 500 code-mixed Malay-English (Manglish) sentences collected from recent Malaysian social media comments on platforms such as Reddit Malaysia, TikTok, Facebook and X, labelled by the team into 274 toxic and 226 non-toxic. This set is used for few-shot fine-tuning by adding a small portion of code-mixed data into training, so that the model becomes more relevant to the everyday toxic and non-toxic context used in Malaysia.

2.3 Data Preprocessing Methods

2.3.1 Text Cleaning & Normalisation

The dataset was cleaned and normalised using a custom function with the design choices based on what carries toxicity signals. Due to the existing placeholder tags from the original dataset, each tag was handled accordingly, where anonymisation tags such as <user> and <number> were removed, while emoticon placeholder tags like <happy>, <wink> were converted into plain words to preserve their original emotional signal. URLs, @mentions, newline characters, repeated and leading whitespace were also removed. The smart quotes were normalised into straight quotes for standardization.

Several parts that carried meaningful toxicity cues and suit XLM-RoBERTa, our subword tokenizer were preserved. These include the original casing that signals aggression like 'SHUT UP', emojis that carry emotional signals, repeated punctuation such as "!!!" that indicates intensity, apostrophes, special characters and symbols. User-typed bracket content like '<3' heart or '< muzik teras >' also remained in the text as these were the real content typed by users.

2.3.2 Data Splitting & Experimental Setup

The cleaned Mendeley dataset were split using a stratified 70/15/15 ratio into training, validation, and test sets. Stratification combined both language and label so that English or Malay and toxic or non-toxic proportions stayed balanced across all three splits. To evaluate whether adding code-mixed data improves Manglish detection, the 500 curated code-mixed sentences were first split in a stratified manner into 300 training, 75 validation, and 125 test sentences. Then, two versions of the training data were prepared. In the zero-shot processed dataset, training and validation remained only monolingual English and Malay, while in the few-shot processed dataset, 300 code-mixed sentences were folded into training and 75 into validation. The remaining 125 code-mixed sentences were held out as a test set and kept identical across both versions, so both models are tested on exactly the same code-mixed Malay and English data.

2.4 Training Procedures

The model was fine-tuned end-to-end using a fully supervised training pipeline executed on a GPU environment. All tunable hyperparameters were centralized in a single configuration block, which are a maximum sequence length of 128, batch size of 16, learning rate $2e-5$, up to 5 epochs with early stopping patience of 3, warmup ratio of 0.1, weight decay of 0.01, dropout rate of 0.3, and a hidden dimension of 256 for the classification head.

The AdamW optimiser was configured with selective weight decay, where the bias terms and LayerNorm weights were placed in a separate parameter group with weight decay set to zero, following the standard BERT fine-tuning convention. A linear warm-up learning rate schedule was applied using HuggingFace's *get_linear_schedule_with_warmup*, ramping the learning rate from 0 to $2e-5$ over the first ten percent of the total training steps before linearly decaying it back to zero, stabilizing early training when the randomly-initialized classification head is far from convergence.

Mixed precision training (*torch.amp.autocast*) was enabled to halve GPU memory usage with negligible accuracy loss, effectively allowing a larger batch throughput on the T4 GPU. Then, Gradient clipping (*max_norm = 1.0*) was applied after unscaling to prevent exploding gradients during the warm-up phase. The scheduler was stepped after every batch rather than every epoch, which is required for the linear warm-up to function correctly.

At the end of each epoch, the model was evaluated on the validation set using binary F1-score as the primary selection metric. If validation F1 improved, the model weights were saved as the best checkpoint. Training will end early if no improvement was observed for three consecutive epochs. The best checkpoint was subsequently restored and uploaded to the HuggingFace Hub for deployment. Two separate training runs were conducted: a zero-shot run on the monolingual English and Malay data only, and a few-shot run with 300 additional Manglish (Malay + English) samples injected into the training set. A fixed random seed of 42 was applied across PyTorch, NumPy, and Python's random module throughout to ensure reproducibility.

The loss function used throughout training was CrossEntropyLoss, which internally applies Softmax over the two output logits, making an explicit Softmax layer in the classification head unnecessary. The GradScaler from torch.amp was paired with the autocast context to prevent underflow in half-precision gradient values during backpropagation. Per-batch loss, accuracy, and binary F1 were aggregated and logged at the epoch level alongside validation metrics, enabling systematic comparison across epochs and between the two experimental runs. The training DataLoader was configured with shuffle=True, num_workers=2, and pin_memory=True to maximise GPU data transfer throughput, while all validation and test loaders ran without shuffling to ensure deterministic evaluation.

2.5 Tools & Frameworks

The system was implemented in Python on Google Colab with GPU acceleration. Core deep learning components relied on PyTorch and Hugging Face Transformers for the XLM-RoBERTa tokeniser and encoder, with scikit-learn for evaluation metrics, Captum for integrated-gradients word-level attribution, and Gradio for the interactive frontend dashboard. Model checkpoints were stored and served via the HuggingFace Hub.

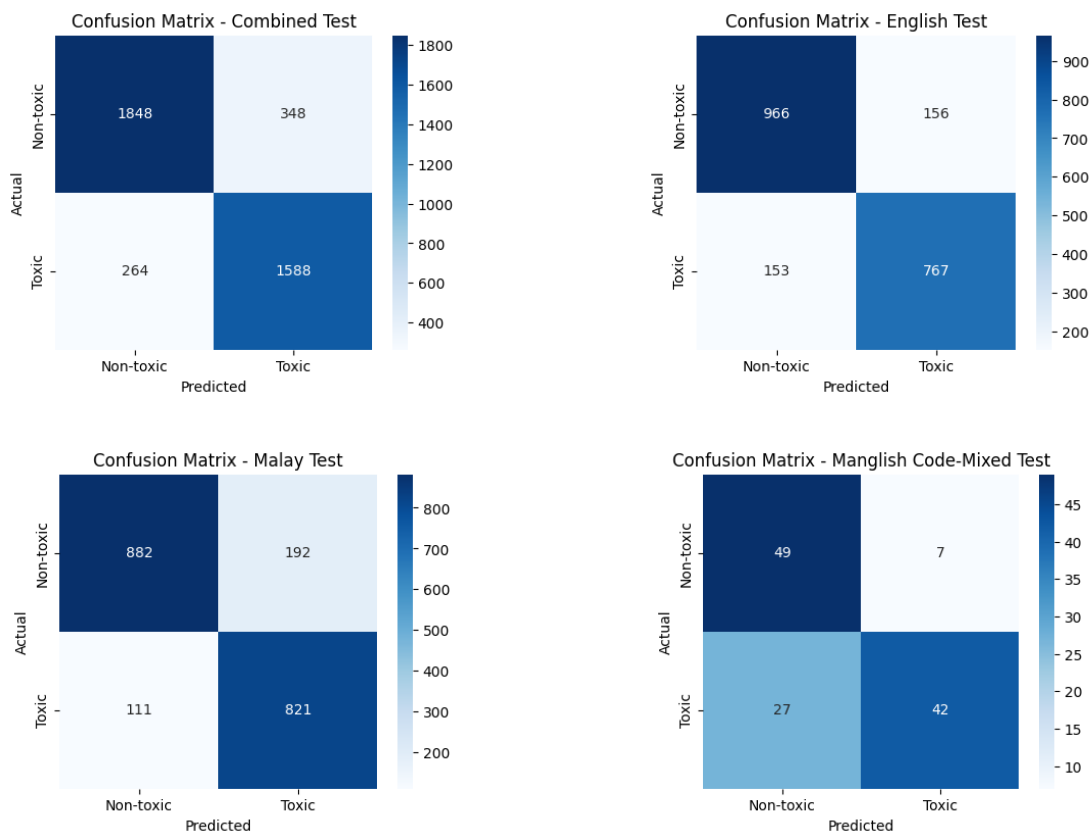
3.0 Results

3.1 Per-Language Results Table

Dataset	Accuracy	Precision	Recall	F1	MCC	AUC-ROC
Combined Test	0.8488	0.8202	0.8575	0.8384	0.6971	0.9258
English Test	0.8487	0.8310	0.8337	0.8323	0.6945	0.9172
Malay Test	0.8490	0.8105	0.8809	0.8442	0.7004	0.9311
Manglish Code-Mixed Test	0.7280	0.8571	0.6087	0.7119	0.4927	0.8080

Table above presents the performance of the few-shot XLM-RoBERTa model across English, Malay, Manglish, and the combined test set. The model achieved strong performance on the monolingual datasets, with both English and Malay recording approximately 84.9% accuracy. The Malay dataset produced the best overall results, achieving the highest F1-score (0.844), recall (0.881), MCC (0.700), and AUC-ROC (0.931), while the English dataset also performed consistently with an F1-score of 0.832. In contrast, the Manglish dataset remained the most challenging dataset, obtaining an F1-score of 0.712 despite achieving the highest precision (0.857), indicating that some toxic code-mixed comments were still missed. Overall, the combined test set achieved an F1-score of 0.838 and an AUC-ROC of 0.926, demonstrating effective multilingual toxic comment detection.

3.2 Confusion Matrices



The confusion matrices provide a detailed view of the model's classification performance across different language settings. English and Malay achieved strong results with high numbers of correctly classified toxic and non-toxic comments, while Malay recorded the fewest false negatives, contributing to its highest recall score. In comparison, Manglish remained the most challenging dataset, as a larger number of toxic comments were misclassified as non-toxic, although the low false positive count indicates that the model remained conservative when assigning toxic labels.

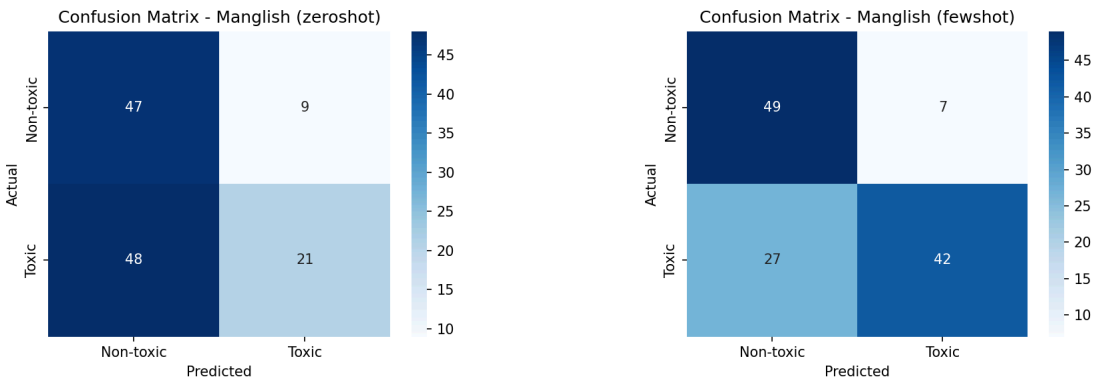
3.3 Effect of Code-Mixed Fine-Tuning

To evaluate whether adding code-mixed data improves Manglish detection, both zero-shot and few-shot models were evaluated on the same test set of 125 Manglish sentences. The evaluation results are as followed:

Model	Accuracy	Precision	Recall	F1	MCC	AUC-ROC
Zero-shot	0.5440	0.7000	0.3043	0.4242	0.1672	0.7089
Few-shot	0.7280	0.8571	0.6087	0.7119	0.4927	0.8080

As shown above, adding just 300 code-mixed sentences to training substantially improved the model's performance on Manglish text. F1 rose from 0.424 to 0.712 and recall from 0.304 to 0.609, while accuracy also improved from 0.544 to 0.728 and precision from 0.700 to 0.857. The stricter MCC metric jumped from 0.167, close to random guessing, to 0.493, showing the few-shot model genuinely learned to tell toxic from non-toxic Manglish, and AUC-ROC rose from 0.709 to 0.808 for more reliable separation across thresholds.

The confusion matrices in figure below illustrate the improvement. The zero-shot model correctly identified only 21 of the 69 toxic Manglish comments and missed 48 of them. After a few-shots fine-tuning on code-mixed data, the model caught 42 of the 69 toxic comments, while keeping false positives low at 7 out of 56 non-toxic comments.



4.0 Evaluation & Discussion

4.1 Comparative Analysis (monolingual vs code-mixed)

A comparison between the monolingual datasets (English and Malay) and the code-mixed Manglish dataset reveals a clear performance gap. English and Malay achieved F1-scores of 0.832 and 0.844 respectively, while Manglish achieved a lower F1-score of 0.712. Similar trends were observed for recall and AUC-ROC, indicating that toxicity detection becomes more challenging when Malay and English are mixed within the same sentence.

The stronger performance on English and Malay is likely due to the substantially larger amount of monolingual training data available, compared to only 300 code-mixed samples used during few-shot fine-tuning. Nevertheless, the few-shot model still achieved an F1-score of 0.712 and an AUC-ROC of 0.808 on the Manglish test set, demonstrating that the multilingual XLM-RoBERTa architecture is capable of adapting to code-mixed Malaysian social media language with limited additional training data. These findings suggest that multilingual transformer models provide strong performance for monolingual toxic comment detection while remaining effective in code-mixed settings.

4.2 Strengths

The implemented system exhibits a robust cross-lingual knowledge transfer capability. By utilizing XLM-RoBERTa as its foundation, the model is able to exploit shared subword representations across its massive vocabulary. This allows the model to map semantically similar terms into a single embedding space. This enables it to capture contextual meaning across English and Malay seamlessly, achieving a highly consistent monolingual accuracy of approximately 84.9% for both domains without requiring separate, language-specific models.

Furthermore, the architecture is highly data-efficient when adapting to localized code-mixed Manglish text. By introducing a targeted few-shot fine-tuning injection of just 300 manually collected Manglish sentences, the model's F1-score on the code-mixed test set increased from a weak zero-shot baseline of 0.424 to 0.712. This proves the encoder's strong capacity to quickly learn and understand real-world Malaysian conversational context with a small amount of data.

Last but not least, the solution is heavily optimized by an information-preserving preprocessing framework and a data-driven configuration. Rather than relying on naive text cleaning that removes all non-alphanumeric data, the pipeline selectively retains structural elements that carry strong semantic meaning. While anonymization tags are removed to protect privacy, aggression markers like all-caps writing, repeated exclamation marks, and emojis are intentionally preserved, and emoticon tags are converted back to plain words to capture more toxicity signals. Combined with an empirical token length analysis that safely sets the sequence length to 128 to prevent training truncation, this tailored pipeline ensures highly precise filtering.

4.3 Error Analysis & Weaknesses

For Manglish comments, the most critical failure in zero-shot setting is recall collapse, where the model achieves F1 of only 0.424 with precision at 0.7 and recall at 0.304. It misses 48 of 69 genuine toxic comments. Most false negatives involve Malay slurs in-grained with English-framed sentences like “My son is good boy. U yg panggil dia char siew boy bodo banget” (toxic_prob=0.024) and “Bodoh apa trying to overtake in what looks like a residential area?? Laju gila pulak tu wtf” (toxic_prob=0.172), which the encoder contextualises as neutral words due to absent code-mixed training signal. Few-shot exposure of 300 Manglish training samples reduces false negatives (missed toxic) to 27, raises F1 to 0.712, confirming a data coverage gap.

English false positives (false alarm) remain consistently elevated across both experiments (FP=152 zero-shot, 156 few-shot, F1 unchanged at 0.832). They are mostly driven by three patterns, which are comments where offensive words used casually among friends rather than attacks, sarcastic remarks where hostile wording is not meant literally and comments that discuss sensitive topics without targeting anyone specifically. None are resolved by few-shot Manglish exposure, indicating the model requires annotated irony and in-group speech examples to reduce this gap.

Few-shot training introduces a Malay precision-recall trade-off, where false positives rise from 139 to 192 as the model overgeneralises Malay-script tokens seen in Manglish toxic sentences to Malay-only inputs, dropping precision from 0.851 to 0.810. Thus, prepending explicit language tags like [EN], [MS] and [CM] during training and inference can be one of the ways to mitigate this cross-contamination.

4.4 Future Improvements

There are several directions that can further improve the system. First, expanding the Manglish training set beyond the current 300 samples through crowdsourcing or data augmentation, would directly address the recall gap observed in the code-mixed detection. Next, prepending explicit language tags such as [EN], [MS], and [CM] during both training and inference could mitigate the cross-language contamination that caused a precision drop in Malay after few-shot fine-tuning. Third, the model currently struggles with sarcasm, casual offensive language among friends, and context-dependent insults, hence incorporating annotated irony and in-group speech examples into training would reduce persistent English false positives. Finally, exploring larger multilingual models such as XLM-RoBERTa-large or fine-tuning on a continually updated stream of social media data could improve the robustness to the rapidly evolving vocabulary of online toxic language.

Beyond data and architecture improvements, the current binary classification of toxic/ non-toxic schemes could be extended into a multi-class classification that distinguished between specific toxicity categories such as hate speech, threats, racism, or personal attacks, providing more actionable signals for content moderators. Additionally, deploying the model in a continual learning framework, where newly flagged comments from real platform traffic are periodically reviewed and folded back into retraining, would allow the system to adapt to emerging slang, evolving toxic patterns, and new social contexts without requiring a full retraining cycle from scratch.